

Feedback and Newcomer Retention in X Community Notes

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Abstract

In X’s Community Notes, a contributor’s first note can be rated Helpful, rated Not Helpful, or left in “Needs More Ratings” (NMR) with no verdict. We ask whether a Not Helpful verdict on a first note is associated with different retention than no verdict. Studies of Wikipedia newcomers cannot make this comparison, because Wikipedia has no equivalent of a contribution that never receives a verdict. Using a public snapshot of 3.1 million notes, we classify each author’s first note by its status seven days after it was written. Authors whose first note had been rated Not Helpful by then were 15.0 percentage points less likely to write another note within 30 days than authors whose note was still in NMR (95% CI -15.5 to -14.5). Most of this difference coincides with a platform rule. Since April 2024, a single Not Helpful note removes a new author’s ability to write, and 89.4% of authors whose first note was rated Not Helpful have lost writing ability, usually on the day the verdict arrived. The difference is 6.1 points for cohorts that started before the rule and 17.1 points for cohorts that started after it. In the pre-rule cohorts, at least 4.9 points of the difference cannot be explained by the lockout.

1 Introduction

Crowdsourced moderation depends on contributors who keep participating after their first attempt. In Community Notes, volunteers write short notes that add context to posts on X, and other volunteers rate those notes. A matrix-factorization model publishes a note only when raters who usually disagree with each other both find it helpful [4]. The bar is high, and most notes never clear it.

As a result, a first note has three possible outcomes: it is rated Helpful, it is rated Not Helpful, or it receives no verdict and stays in Needs More Ratings (NMR) indefinitely. Research on Wikipedia has shown that negative early feedback predicts newcomer departure: newcomers whose first edits are reverted are less likely to keep editing [1, 2]. On Wikipedia, however, an edit that is not reverted stays in the article, so there is no large group of newcomers whose first contribution received no response. In Community Notes, no response is the most common outcome.

Our question is whether an explicit rejection (Not Helpful) is associated with different retention than no verdict (NMR). Helpful serves as a reference point. The question matters for design. A system in which most contributions never receive a verdict has in effect chosen a default response to newcomers, and it is not obvious whether that response is better or worse for retention than a rejection.

We find that a Not Helpful first note predicts much lower retention than an NMR first note. A large part of this difference comes from a platform rule we did not know about when we began: since

April 2024, a Not Helpful first note automatically removes the author’s ability to write notes. Section 6.2 separates the part of the difference that this rule accounts for from the rest. In cohorts that started before the rule, a difference of about five points remains.

2 Background

Two platform rules matter for interpreting our results.

Contributors must rate before they can write. New contributors can only rate notes. They gain the ability to write when their Rating Impact reaches 5. Rating Impact increases when a contributor rates a note before it receives a status and the rating matches the status the note later receives [6]. This requirement was introduced in September 2022 and applies to every cohort in our analysis. The authors we call newcomers are therefore new to writing only. Each of them has already rated enough notes, early and accurately enough, to pass this threshold. They are probably more committed than a typical first-time Wikipedia editor, which weakens the comparison to that literature.

A Not Helpful verdict can remove the ability to write. A contributor loses writing ability when three of their last five resolved notes are rated Not Helpful. Since 2 April 2024, they also lose it when any of their notes is rated Not Helpful while their Writing Impact is zero or negative [6]. Writing Impact starts at zero and changes only when a note resolves, so a first-time author’s Writing Impact is exactly zero. From April 2024 onward, a single Not Helpful first note is therefore enough to remove writing ability, while an NMR first note has no consequence. In other words, the platform itself treats the two outcomes we compare differently. We return to this rule in Section 6.2.

Statuses are recomputed periodically and can change, but the public data records when each note first left NMR and which status it received [7]. We use that first transition.

3 Related Work

The most relevant prior work studies how Wikipedia retains its newcomers. Halfaker et al. [2] analyzed 400,000 revisions and found that having early edits reverted strongly reduces how much newcomers contribute afterward, although reverting also improves the overall quality of the encyclopedia. A later study [1] attributes much of Wikipedia’s decline to the increasingly automated and impersonal rejection of newcomers’ edits.

Community Notes differs from Wikipedia in two ways that matter for our question. First, rejection comes from a scoring algorithm rather than from another editor, so a rejected author has no one to respond to and no explanation to read. Second, the most common outcome is no verdict at all, which Wikipedia does not produce at a comparable scale. Mohammadi et al. [3] describe the program’s first four years, including contributor activity and rating behavior.

To our knowledge, no prior work has examined how the absence of a verdict relates to retention.

4 Data

We use the public Community Notes data downloads [5], from the snapshot released on **18 September 2026**. Each snapshot covers all notes back to 2021, so one snapshot is sufficient. The download site keeps only about a week of snapshots, so our local copy is the only one available to us, and our pipeline never modifies it.

The snapshot contains three tables: notes (3,086,867 rows), noteStatusHistory (3,290,294 rows), and userEnrollment (1,490,672 rows). Our analysis relies mainly on two fields in noteStatusHistory: firstNonNMRStatus, the first status other than NMR that a note received, and timestampMillisOfFirstNonNMRStatus, the time it received that status. Both fields are empty for notes that never left NMR, which is the case for 82.7% of notes.

A snapshot includes only notes created at least 48 hours before its release, so our data ends on **16 September 2026**. We use this cutoff, not the snapshot date, for all censoring.

We did not use the ratings table, which is tens of gigabytes and records individual ratings rather than note outcomes. We also did not use the text of the posts being annotated, which is not publicly available.

Stability of author identifiers. Our analysis assumes that an author identifier refers to the same person in every release. Because snapshots expire, this can only be checked while a second snapshot is still available. We compared noteStatusHistory in the 18 September and 16 September releases. Of 3,282,773 notes present in both, none had a different author identifier.

Data quality problems. We found two problems in the data. First, firstNonNMRStatus is empty for every note created before May 2022 (28,988 notes). Cohorts from that period appear to be 100% NMR because verdicts were not recorded, not because the notes went unresolved. Second, 233,514 notes (7.6%) have no row in noteStatusHistory. These notes span 28 months starting in March 2024, and their share of each month’s notes rises from 6.7% to 14.1% over that period. The outcome of these notes is unknown, which is different from never leaving NMR. Counting them as NMR would add a growing number of notes to the comparison group and create a spurious trend across cohorts. Both problems lead to exclusions, described in Section 5.4.

5 Method

5.1 Contributor table

We join notes to noteStatusHistory on note ID and build a table with one row per author. Each row records the author’s first note, its outcome and when the outcome arrived, the time of the author’s next note, the month of the first note (the author’s cohort), and features of the first note: its classification, whether it cites a URL, its length, and its reason tags. An author’s first note is their earliest note in the snapshot, with ties broken by note ID so that the ordering is deterministic. Because the snapshot goes back to 2021, this is the author’s actual first note. The table has 343,176 rows, one for each distinct author in notes.

We validated the table against the raw files with twelve structural checks. We also chose twenty authors at random and rebuilt each of their rows from the raw TSV files with a separate Python parser that does not use our query engine. All twelve checks passed, and all twenty rows matched.

5.2 Landmark classification

Our most important methodological choice is to classify each first note by **its status seven days after it was written**, rather than by the status it eventually received.

A note’s eventual status can reflect events that had not happened when the author was deciding whether to write again. A note that stayed in NMR for three months before being rated Not Helpful had no verdict during the period when its author was making that decision. In addition, whether a note is still in NMR at the end of the data depends partly on when the snapshot was taken.

Seven days is enough time for nearly all verdicts to arrive. For first notes written in 2023 or later, the median time to a verdict is under eight hours, and the 90th percentile is under two days. We exclude authors who wrote a second note before day seven, because they returned before the point at which we classify outcomes, so their return cannot be a response to the day-7 status. Notes that receive a verdict after day seven are classified as NMR.

5.3 Retention and censoring

An author counts as retained if they wrote another note between day seven and the end of a 30-, 60-, or 90-day window. An author is included in a window only if the whole window ends before the data cutoff, so the sample size differs across windows. We report proportions with Wilson 95% confidence intervals and differences with Newcombe hybrid-score intervals. To check that this windowing does not drive the results, Section 6.3 also reports Kaplan–Meier estimates, which censor incomplete observations instead of dropping them.

5.4 Exclusions

Each exclusion is a boolean column applied by a single function, so no rows are deleted. Of 343,176 authors, 322,094 remain after the following exclusions:

- 5,052 authors whose first note was written before January 2023. For these cohorts, verdicts are either missing or were changed when X rescored all historical notes in January 2023.
- 15,997 authors whose first note has no status-history row, so its outcome is unknown.
- 35 AI note-writer accounts, identified by an enrollment state beginning with api. These accounts are few but wrote 8.6% of all notes, and an automated account that stops writing does not tell us anything about people leaving.

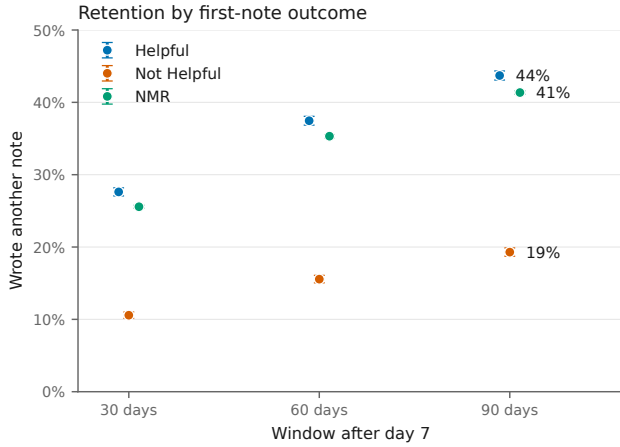
Applying the day-7 landmark leaves 230,588 authors, of whom 227,324 have a complete 30-day window.

6 Results

Authors whose first note was rated Not Helpful within seven days were 15.0 percentage points less likely than authors whose first

Table 1: Retention by first-note outcome at day 7. Wilson 95% intervals.

First-note outcome	<i>n</i>	30-day	90-day
Helpful	24,798	27.6%	43.7%
Not Helpful	19,153	10.6%	19.3%
No verdict (NMR)	183,373	25.6%	41.4%

**Figure 1: Retention by first-note outcome at 30, 60, and 90 days, with 95% intervals.**

note was still in NMR to write another note in the following 30 days (95% CI -15.5 to -14.5 ; $n = 19,153$ Not Helpful and 183,373 NMR). Section 6.2 shows that most of this difference coincides with the writing lockdown.

6.1 Not Helpful versus no verdict

Table 1 and Figure 1 show that the two outcomes differ from NMR by very different amounts. A Not Helpful first note is associated with much lower retention than an NMR first note: 15.0 points lower at 30 days and 22.1 points lower at 90 days. A Helpful first note is associated with only slightly higher retention than an NMR first note: 2.1 points higher at 30 days and 2.4 points higher at 90 days. Authors who received no verdict returned at nearly the same rate as authors whose notes were rated Helpful.

A simpler analysis gives a larger difference. If each note is labeled with its eventual status and retention is measured from the day the note was written, the difference is 21.1 points instead of 15.0, about 40% larger. Most of the extra difference comes from including authors who wrote again within the first week. These are the most active authors in the data, and they returned before the point at which we classify outcomes. Both analyses use the same censoring rules, so the gap between them comes from the landmark alone.

6.2 Cohorts and the April 2024 lockdown rule

Figure 2 shows the difference by cohort month. Through 2023 and early 2024, the difference is small and often not distinguishable from zero. It then widens sharply. Thirty-day retention for Not Helpful

authors falls from about 28% in February 2024 to 9.6% in April 2024 and stays between 3% and 8% afterward, while retention for the other two groups declines together and much more gradually. Pooling the cohorts on each side of April 2024, the Not Helpful minus NMR difference is -6.1 points before (95% CI -7.4 to -4.7) and -17.1 points after (-17.6 to -16.7).

April 2024 is when X began removing writing ability after a single Not Helpful note (Section 2). The enrollment data shows the rule in operation. Of authors whose first note was rated Not Helpful, 89.4% have lost writing ability at least once, compared with 16.8% of authors whose first note stayed in NMR, and 48.7% are still unable to write. Among Not Helpful authors who wrote only one note and later lost writing ability, the median time between the verdict and the loss of writing ability is 0.03 days (under an hour), and 84.0% lost it within a week. For most of these authors, the verdict and the loss of writing ability happened at essentially the same time.

The lockdown does not explain the whole difference. The enrollment table records only a contributor’s most recent loss of writing ability, so an author who started in 2023 may have lost it years later under the new rule. Counting only losses recorded before 2 April 2024, 1.2% of pre-rule Not Helpful authors had lost writing ability during the period we study, compared with 0.3% of pre-rule NMR authors. Nearly all pre-rule authors were still able to write, and their difference is still 6.1 points. Even if every author who lost writing ability would otherwise have returned, the lockdown can account for at most 1.2 points of that difference, which leaves at least 4.9 points that it does not explain.

We therefore report two estimates that answer different questions. The pooled difference of 15.0 points describes what happens today to newcomers whose first note is rated Not Helpful, and most of it reflects the lockdown rule. The difference of 4.9 points in pre-rule cohorts is our estimate of the association between rejection and not returning when the lockdown does not apply.

6.3 Time to return

Kaplan–Meier estimates agree with the windowed proportions, so the windowing is not driving the results. Follow-up starts at day seven, the event is the author’s next note, and authors who have not written again are censored at the data cutoff instead of being dropped. By day 90, 43.7% of Helpful authors, 41.2% of NMR authors, and 19.0% of Not Helpful authors had written another note, all within half a point of the windowed estimates in Table 1. Because fewer than half of the authors in every group return within 90 days, no group reaches a median time to return in that period.

6.4 Adjusted estimates

We also fit a logistic regression of 30-day retention on day-7 status, with fixed effects for cohort month and controls for features of the first note (classification, URL presence, length, and reason tags). The odds ratio for Not Helpful is 0.36 (95% CI 0.34–0.38), which corresponds to an average marginal change in predicted retention of -18.5 points. The adjusted estimate is 3.5 points larger than the unadjusted difference, and we report both. Among the controls, a first note stating that a post is *not* misleading is associated with 2.4 points higher retention when status is held constant, even though

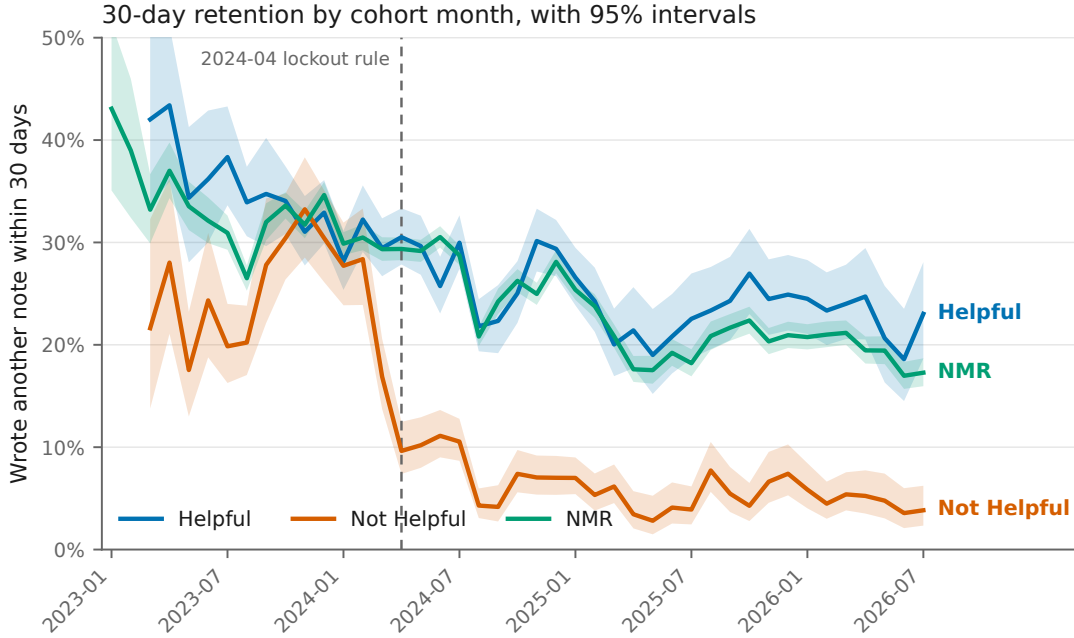


Figure 2: 30-day retention by cohort month and first-note outcome, with 95% intervals. The dashed line marks April 2024, when a single Not Helpful note began to remove writing ability. Cohort months too recent for a 30-day window to close before the data cutoff are omitted.

Table 2: Not Helpful minus NMR difference at 30 days under each variation.

Variation	Difference
Baseline (day-7 landmark)	−15.0 pp
Landmark at 3 days	−18.2 pp
Landmark at 14 days	−11.7 pp
Excluding not-misleading first notes	−14.0 pp
Retention = two or more later notes	−5.7 pp
Cohorts with 90+ days after the window	−15.0 pp
Eventual-status specification	−21.1 pp

such notes cannot be rated Helpful and are rated Not Helpful much more often.

7 Robustness

Table 2 repeats the main comparison under five alternative specifications. The Not Helpful minus NMR difference is negative in every case, ranging from −5.7 to −18.2 points, compared with −15.0 in the baseline. Moving the landmark changes the estimate in the expected direction. An earlier landmark keeps more authors in the sample but classifies as NMR some notes that later receive a verdict, and a later landmark does the reverse. Requiring two or more later notes reduces the difference because it lowers retention in every group. None of these changes is as large as the difference between the pre-rule and post-rule cohorts.

8 Threats to Validity

Writing lockouts. This is the largest threat to our interpretation. Section 6.2 bounds its contribution in pre-rule cohorts, but after April 2024 we cannot separate it from other reasons for leaving. `userEnrollment` records only each contributor’s current state, with no history. We therefore cannot distinguish an author who was unable to write when they would have returned from one who had already decided to stop. An author who lost writing ability, regained it, and then left looks the same as one who never lost it. The pre-rule estimate avoids this problem by using a period when the rule did not exist, at the cost of a smaller and older sample.

Selection. We did not assign authors to outcomes. A weak first note may both be rated Not Helpful and come from an author who was likely to leave anyway. Controlling for observable features of the note does not resolve this, so we describe all results as associations.

Notification. We cannot observe whether an author saw their note’s status. The landmark assumes that seven days is enough time for an author to learn of a verdict, which depends on X’s notifications and cannot be verified from public data.

Survivorship. Deleted notes are removed from notes but remain in `noteStatusHistory`, so our authors are drawn only from notes that were not deleted.

Automated authors. AI note writers joined in 2025. We exclude them based on their current enrollment state, so an AI account that has since moved to a different state would be missed.

9 What We Did

Most of our four weeks went into understanding the system and the data rather than into the statistical analysis.

In the first week we worked out how the system operates. Reading the scoring source code alongside the published guide is how we found the April 2024 lockout rule and the date it took effect. Knowing the date is what made the before-and-after comparison in Section 6.2 possible; without it, we could only have listed the lockout as a confound. The guide’s web pages are rendered client-side and return no content to a plain HTTP request, so we worked from the Markdown files the site is built from.

In the second week we built the contributor table and found the two data problems described in Section 4. Verdict fields are empty before May 2022, which ruled out the 2021 and 2022 cohorts. And 7.6% of notes have no status-history row, which is not documented and would have added a growing number of unknown outcomes to the NMR group. We stopped to investigate each problem before continuing, which took about a day each. The same week, we checked that author identifiers were stable across two releases, which is only possible while both releases can still be downloaded.

Several infrastructure problems also took time. Our Python environment is a pinned Nix flake. Our first attempt used `uv`, which failed because its downloaded Python interpreter is dynamically linked against a loader that does not exist on NixOS. The survival-analysis library we planned to use did not build with the pinned packages, so we computed the Kaplan–Meier estimates and the log-rank test with `statsmodels`. Reading the tables with default CSV quoting silently corrupted rows, because some note text contains unescaped double quotes.

We also caught two of our own mistakes. The first version of the cohort figure showed a spike in every series in the most recent month. Only authors who joined in the first few days of that month had a 30-day window that closed before the cutoff, so that month’s estimate came from a biased subset of authors. We fixed this in the analysis with an explicit completeness flag rather than by trimming the chart. Second, our first draft of the regression results said the adjusted and unadjusted estimates agreed, when they differ by 3.5 points.

We kept the pipeline reproducible throughout. The contributor table was frozen after validation and never rebuilt, all exclusions are applied by one function, and the numbers in this report come from files generated by the analysis scripts rather than from manual calculation.

10 Conclusion

A Not Helpful verdict on a first note predicts much lower continued contribution than no verdict: 15.0 points lower at 30 days. Most of this difference coincides with X’s writing lockout. In cohorts that started before the lockout rule, when a rejection did not remove the ability to write, the difference is about five points. Rejection on its own is associated with a modest drop in retention, and the lockout accounts for most of the rest of the pooled difference.

We did not expect how little a Helpful verdict adds. Authors whose first note was rated Helpful returned only about two points more often than authors who received no verdict. This has two implications for a system in which 82.7% of notes never receive

a verdict. Receiving no verdict is not associated with the drop in retention that follows a rejection, so the large share of unresolved notes does not appear to be costing the system contributors. But a Helpful verdict is also associated with little additional retention. Our results therefore give little reason to expect faster verdicts to retain more new authors. A more relevant design question is whether a first rejection should remove the ability to write at all, given that 89.4% of authors whose first note is rated Not Helpful lose that ability, usually on the same day.

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